Koushik Kampa

Senior Data Engineer | Multi-Cloud Platforms | Streaming & AI/ML Data Engineering

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SUMMARY

Senior Data Engineer with **11+ years** designing, building, and scaling multi-cloud data platforms across **GCP, AWS, Azure, and Confluent Cloud**. Expert in real-time and batch data engineering with **Apache Spark, PySpark, Apache Beam/Dataflow, Flink, Kafka/Confluent, Kafka Connect, Debezium**, and modern lakehouse architectures (**Apache Iceberg, Delta Lake**). Strong in **ETL/ELT, dbt (tests/docs), Airbyte, Fivetran, CDC, data contracts**, and end-to-end quality/observability using **Great Expectations, Soda, OpenLineage, Purview, DataHub, Unity Catalog, Monte Carlo**. Programming: **Python, SQL, Scala**. IaC/DevOps: **Terraform, Docker, Kubernetes, Jenkins, GitHub Actions, Cloud Build**. Delivered **40% faster queries**, **20–30% cloud cost reduction**, and **35% higher platform uptime** via **partitioning, clustering, BigQuery Reservations, Redshift Spectrum, Synapse Serverless**, and right-sizing. Recent focus on **AI/ML-ready engineering—feature stores**, **embedding pipelines**, **vector databases (Milvus, Qdrant, pgvector)**, and **RAG** to power LLMs and advanced analytics. Domain coverage: **Telecom, Financial Services, Healthcare, Retail/E-commerce, Cloud AI**, delivering secure, governed, and compliant solutions (**HIPAA, GDPR, PCI DSS, SOX**). Known as a **mentor** and **Agile lead**, partnering with **Product, Compliance, and DevOps** to drive **DataOps/MLOps** and **data mesh/data product** practices.

CAREER HIGHLIGHTS

- **Engineered multi-cloud data platforms** across GCP, AWS, Azure, and Confluent Cloud, integrating 10+ enterprise sources (Teradata, Oracle, SQL Server, PostgreSQL, Snowflake) into BigQuery, Redshift, and Synapse for unified analytics.
- **Designed high-volume streaming pipelines** with Kafka, Confluent Cloud, Redpanda, Flink, and Spark, processing 500M+ events/day with sub-second latency to power fraud detection, personalization, and real-time insights.
- **Implemented modern lakehouse architectures** (Iceberg, Delta Lake) with schema evolution, time travel, and cross-engine querying, enabling scalable analytics and AI/ML workloads while delivering 20–30% cloud cost savings.
- **Established data governance, security, and quality frameworks** using dbt, Great Expectations, Soda, OpenLineage, Purview, Unity Catalog, and Monte Carlo, ensuring HIPAA, GDPR, PCI DSS, and SOX compliance with audit-ready lineage.
- **Delivered AI/ML-ready pipelines and automation** (RAG, embeddings, vector databases, feature stores) alongside IaC/CI/CD with Terraform, Kubernetes, Jenkins, and GitHub Actions, cutting release cycles by 40%, boosting uptime by 35%, and enabling faster deployment of LLM and analytics use cases.

TECHNICAL SKILLS

GCP: BigQuery, **BigTable**, **Dataflow** (Beam), Pub/Sub, **DataPlex**, Cloud Storage, **Dataproc**, Data Fusion, **Cloud SQL**, Cloud Run, Cloud Composer, Cloud Functions, **Memorystore**, IAM, GKE, Cloud Monitoring/Logging

AWS: S3, **Athena**, **Redshift**, **Glue** (Jobs & Data Catalog), **EMR**, **Kinesis** (Streams/Firehose), MSK, Step Functions, **Lambda**, **DynamoDB**, Lake Formation, CloudWatch, CloudTrail, **IAM**

Azure: ADLS Gen2, Synapse Analytics (Dedicated/Serverless SQL), **Data Factory** (ADF), **Databricks**, Event Hubs, Stream Analytics, Purview, Key Vault, **RBAC**, Microsoft Fabric/OneLake (working knowledge)

Confluent Cloud: event streaming, schema registry, **CDC** integration

Big Data & Streaming: Spark, **PySpark**, **Flink**, **Kafka**, Kafka Connect, Debezium, Confluent, **Redpanda**, **Airbyte**, **Fivetran**, Airflow, Oozie, **Hadoop** (**HDFS**, MapReduce, Hive, Sqoop), Autosys

ETL / ELT & Modeling: dbt (models/tests/docs), Dataflow, Composer, PySpark, **Star/Snowflake schemas**, dimensional modeling, performance tuning, **Data Mesh & Data Products** (working knowledge)

Lakehouse & Formats: Apache Iceberg, Delta Lake, **Parquet**, ORC, **Avro**

Data Quality & Lineage: Great Expectations, Soda, **data contracts** (Avro/Protobuf/JSON Schema), OpenLineage, Purview, DataHub, Unity Catalog (working knowledge), Monte Carlo (working knowledge)

AI/ML Data Engineering: Feature stores, **RAG** pipelines, embeddings, vector DBs (Milvus, Qdrant, pgvector), hybrid search, feature engineering, **MLOps/DataOps** (working knowledge)

Programming: Python (Pandas, NumPy, Polars), SQL, PL/SQL, Scala, Java, Bash/Shell, DuckDB, **pytest**

DevOps, IaC & Delivery: Git, Bitbucket**, Jenkins, GitHub Actions, Maven,** Gradle**, CI/CD, Docker, Kubernetes, Terraform,** CloudFormation**, Agile, Scrum, SDLC, UAT, JIRA, Confluence.**

Security & Cost: RBAC/IAM, VPC-SC (GCP), KMS/Secret Manager, DLP/PII controls, **BigQuery Reservations/Slots**, partitioning & clustering, **HIPAA/GDPR/PCI DSS/SOX** compliance

Visualization & Monitoring: Power BI, **Tableau**, Looker Studio, **Grafana**, Kibana, Stackdriver/Cloud Monitoring, **AWS CloudWatch**, **Azure Monitor**

PROFESSIONAL EXPERIENCE

Client: Chewy — Dec 2023 – Present.

Role: Senior AI/ML Data Engineer

Responsibilities:

- Built **personalization/recommendation** pipelines using customer profiles, orders, and behavior to improve engagement and retention.
- Designed **multi-cloud** pipelines across **AWS** (S3, Glue, Redshift, EMR) and **Azure** (ADLS, Synapse, Databricks) with **Airbyte/Fivetran** for ingestion and CDC.
- Implemented **A/B testing/experimentation** in BigQuery, Redshift, and Synapse to measure uplift and optimize merchandising strategies.
- Developed ETL/ELT with **Spark, PySpark, dbt**, producing analytics-ready feature datasets for ML.
- Used **DuckDB** and **Polars** for local analytics/feature engineering, reducing experimentation cycle time by **30%**.
- Landed Kafka/Redpanda streams into **Iceberg/Delta** tables on S3/ADLS for low-latency personalization and demand forecasting.
- Partnered with data science to operationalize **feature stores**, cutting time to production by **25%**; added **pytest** and dbt tests for regression coverage.
- Established data contracts, validations, and lineage with **Great Expectations, Soda, OpenLineage**; applied **GDPR/CCPA** controls to customer data.
- Automated **IaC/CI/CD** with Terraform, Kubernetes, Jenkins for reliable, repeatable releases.
- Built Looker/Power BI supply chain dashboards, reducing reporting costs by **20%** and improving promotion/fulfillment visibility.

Environment: AWS (S3, Glue, Redshift, EMR), Azure (ADLS Gen2, Synapse, Databricks), GCP (BigQuery), Airbyte, Fivetran, Kafka, Redpanda, Apache Spark, PySpark, dbt, Iceberg, Delta Lake, DuckDB, Polars, Python, SQL, Terraform, Kubernetes, Jenkins, Looker, Power BI, Great Expectations, Soda, OpenLineage, GDPR/CCPA compliance.

Client: Confluent — Mar 2021 – Nov 2023

Role: Senior Streaming & AI Data Engineer

Responsibilities:

- Built real-time streaming pipelines on **Confluent Cloud** and Kafka with **sub-second latency** for high-velocity transactions/events.
- Implemented **CDC** using **Debezium + Kafka Connect** from PostgreSQL/MySQL into **BigQuery** and **Snowflake**.
- Designed streaming **ETL/ELT** with **Flink** and **Spark Structured Streaming** to create curated ML datasets.
- Stood up **Iceberg** tables on S3/GCS with schema evolution, time travel, and multi-engine querying.
- Enforced **data contracts** (Avro/Protobuf) with Schema Registry to prevent breaking changes.
- Embedded **Great Expectations/Soda** for inline quality checks and anomaly detection.
- Delivered lineage/observability with **OpenLineage, Marquez, Monte Carlo**, enabling root-cause analysis and SLA reporting.
- Built **LLM-ready** pipelines integrating streams with **vector DBs** (Qdrant, pgvector) for embeddings, hybrid search, and **RAG**.
- Automated deployments with **Terraform, Kubernetes, GitHub Actions**, reducing release cycles by **40%**; **mentored** teammates on streaming best practices.
- Partnered with platform/data science teams to deliver **feature stores** and real-time experimentation, accelerating personalization and decision-making.

Environment: Confluent Cloud, Apache Kafka, Debezium, Kafka Connect, Apache Flink, Spark Structured Streaming, Iceberg (S3/GCS), BigQuery, Snowflake, Avro, Protobuf, Schema Registry, Python, SQL, Terraform, Kubernetes, GitHub Actions, Great Expectations, Soda, OpenLineage, Marquez, Monte Carlo, GDPR/SOX compliance.

Client: Verizon — Oct 2018 – Feb 2021

Role: Senior Data Engineer

Responsibilities:

- Built real-time and batch pipelines on **GCP (Dataflow/Beam, Pub/Sub, Kafka)** processing over 500M+ records/day into BigQuery and Cloud Storage.
- Optimized **BigQuery, Dataproc (Spark/PySpark), and Cloud Composer (Airflow)** pipelines, reducing failures by 35% and improving query performance by 25–40% through partitioning, clustering, and materialized views.
- Developed **dbt models** with embedded tests and documentation for governed BigQuery transformations, ensuring accuracy and audit-ready lineage.
- Migrated enterprise datasets from **Teradata, Oracle, SQL Server, MySQL, DB2, PostgreSQL, Snowflake, and HBase** into GCP with SLA compliance and automated validations.
- Automated infrastructure provisioning with **Terraform, Cloud Build, and Artifact Registry**; supported capacity planning

and cloud cost optimization with measurable savings.

- Built Python-based validation frameworks integrated with **Cloud Logging and Stackdriver**, detecting anomalies, schema drift, and automating SLA alerts.
- Produced **feature-ready datasets in BigQuery and Spark** to enable ML workflows, including churn prediction and network optimization.
- Secured telecom data pipelines with **KMS/Secret Manager, IAM, and tokenization**, ensuring GDPR and SOX compliance.
- Implemented observability with **Stackdriver, Cloud Monitoring, Grafana, and Looker**, cutting incident response time by 20%.
- Delivered executive dashboards in **Power BI, Tableau, and Looker** for network performance, SLA compliance, and churn insights; mentored junior engineers and led Agile ceremonies.

Environment: GCP (Dataflow/Beam, BigQuery, Cloud Storage, Pub/Sub, Dataproc, Composer/Airflow), Apache Spark, PySpark, Kafka, dbt, Python, SQL, Terraform, Cloud Build, Artifact Registry, Stackdriver, Grafana, Looker, Power BI, Tableau, KMS, Secret Manager, IAM, GDPR/SOX compliance.

Client: BNY — Jul 2015 – Sep 2018

Role: Data Engineer

Responsibilities:

- Developed **Spark/PySpark ETL pipelines on Dataproc**, handling 1B+ records/day and transforming structured/semi-structured data into BigQuery for analytics.
- Tuned **petabyte-scale BigQuery and Dataflow workflows**, reducing execution times by 20–30% via clustering, partitioning, and schema optimization.
- Integrated **Kafka and Pub/Sub** for real-time banking ingestion, enabling near real-time fraud detection with 30% lower latency.
- Migrated enterprise datasets from **Teradata, Oracle, and other RDBMS** into BigQuery using PySpark and modern ingestion tools (including **Debezium, Airbyte, Fivetran**) with CDC, schema alignment, and SLA compliance.
- Automated orchestration with **Cloud Composer (Airflow)**, improving pipeline reliability by 30% and reducing manual intervention.
- Implemented data validation and regression testing in **Python with GCP Monitoring**; added pytest-based unit/integration tests for PySpark and SQL pipelines, reducing incidents by 25%.
- Curated **feature-ready datasets in BigQuery and Spark** to support machine learning and risk analytics, enabling fraud detection and forecasting models.
- Enforced governance using **IAM policies and medallion architecture (bronze/silver/gold layers)** across BigQuery, BigTable, and Cloud Storage, ensuring PCI DSS and SOX compliance.
- Advanced **DataOps/MLOps practices** by integrating CI/CD pipelines with model-feature workflows and automated environment promotion.
- Partnered with **Compliance and Risk teams** to align pipelines with GDPR and audit requirements; contributed to the enterprise data product roadmap within a data mesh approach.

Environment: GCP (BigQuery, Dataproc, Dataflow, Composer/Airflow, Cloud Storage, BigTable), Apache Spark, PySpark, Kafka, Pub/Sub, Teradata, Oracle, RDBMS, Debezium, Airbyte, Fivetran, Python, SQL, pytest, IAM, medallion architecture, Terraform, Jenkins, Bitbucket, PCI DSS, SOX, GDPR compliance.

Client: Clover Health — Apr 2013 – Jun 2015

Role: Data Engineer

Responsibilities:

- Built hybrid-cloud pipelines with **Hive, Sqoop, Spark, Python** for clinical/claims ingestion and transformation.
- Migrated MapReduce ETL to **PySpark/Spark Streaming**, improving throughput by **50%** and maintainability.
- Developed **Kafka + Spark Streaming** with Oracle sinks, reducing manual claims processing by over **150+ hours/month**.
- Designed **HBase** schemas/APIs for patient/provider time series with high write throughput and low-latency reads.
- Delivered **PHI-compliant** datasets for predictive health models; implemented **HIPAA** controls and automated audit checks.
- Authored SQL/Hive/Snowflake (joins, windows, pivots, CTEs) for regulatory and compliance reporting.
- Tuned Spark/Hive jobs, increasing throughput by **30%** and accelerating provider network analytics.
- Built ingestion to Hadoop/Hive via **Sqoop**; created dimensional models (star/snowflake) and stored procedures.
- Added **Great Expectations** and data contracts within PySpark pipelines for continuous validation.
- Produced **Power BI** dashboards with RLS for provider KPIs, ACO metrics, and compliance; **mentored** analysts on SQL and data modeling.

Environment: Hadoop (Hive, Sqoop, MapReduce), Apache Spark, PySpark, Spark Streaming, Kafka, HBase, Oracle, SQL Server, Snowflake, Python, SQL, Great Expectations, Power BI, HiveQL, Dimensional Modeling (Star/Snowflake), PHI data controls, and HIPAA compliance.

EDUCATION

- **Master of Science (M.S.) in Computer Science.**
  Lewis University, Romeoville, Illinois — December 2013
- **Bachelor of Technology (B.Tech.) in Computer Science and Engineering.**
  Jawaharlal Nehru Technological University, Hyderabad (JNTUH) — May 2012

CERTIFICATION

IBM DATA ENGINEERING PROFESSIONAL CERTIFICATE

- Gained in-demand skills in **Python, SQL, and database management** for modern data engineering roles. Completed extensive hands-on projects involving **RDBMS, NoSQL databases, ETL processes, data warehousing, Big Data, and Apache Spark**. Developed practical expertise in **Python scripting, Linux/UNIX shell automation**, and **data pipeline design** for both structured and unstructured data. Also explored **generative AI tools and techniques** relevant to data engineering. Earned an official **IBM digital badge** recognizing technical proficiency and applied project experience.